

Varactor Sizing & Characterization Flow

1. Testbench 1: V_{dc} Sweep Characterization

Testbench: Tese_Varactor_characterization_tb1

Sweep: Parametric sweep of V_{dc} from 0.1 V to 0.6 V (30 linear points).

1.1 Minimum Frequency ($f_{min} = 6$ GHz) / C_{max} Analysis

OUTPUT	EXPRESSION
I_{dev} (Device_current_M7)	i(/V24/PLUS)
$C_{max,meas}$	abs(imag(Device_current_M7) / (2 * 3.14159 * 6e9))
$C_{max,cont}$	flatten(Cap_max_measured, Vdc)
$\frac{dC_{max}}{dV}$	derivative(Cap_max_measured_cont)
$V_{dc,opt}(C_{max})$	x_at_ymax(o(Deriv_Cap_max_measured, analysisName="ac"), 0.1, 0.6, 1)
$\Delta V_{dc}(C_{max})$	xmax(Cap_max_measured_cont) - xmin(Cap_max_measured_cont)
$\Delta C(C_{max})$	ymax(Cap_max_measured_cont, 0.1, 0.6) - ymin(Cap_max_measured_cont, 0.1, 0.6)
$d_{CV}(C_{max})$	Delta_C_cap_max / Delta_Vdc_cap_max
F_{min}	1 / (2 * 3.1416 * sqrt(236.20p * (Cap_max_measured + 1000.07f + 114.5f)))
$F_{min,cont}$	flatten(Freq_min, Vdc)
K_{VCO} at F_{min}	abs(derivative(Freq_min_cont))
$K_{VCO,max}$ at F_{min}	ymax(o(KVCO_at_minF, analysisName="ac"), 0.1, 0.6)
$C_{std,max}$	Cap_max_measured_cont / yvalue(Cap_max_measured_cont, deriv_Vdc_opt_cap_max)
% Error $C_{std,max}$	(ymax(C_standard_cap_max, 0.1, 0.6) - ymin(C_standard_cap_max, 0.1, 0.6)) * 100

1.2 Maximum Frequency ($f_{max} = 9.5$ GHz) / C_{min} Analysis

OUTPUT	EXPRESSION
I_{dev} (Device_current_M7)	i(/V24/PLUS)
$C_{min,meas}$	abs(imag(Device_current_M7) / (2 * 3.14159 * 9.5e9))
$C_{min,cont}$	flatten(Cap_min_measured, Vdc)

OUTPUT	EXPRESSION
$\frac{dC_{\min}}{dV}$	derivative(Cap_min_measured_cont)
$V_{dc,opt}(C_{\min})$	x_at_ymax(o(Deriv_Cap_min_measured, analysisName="ac"), 0.1, 0.6, 1)
$\Delta V_{dc}(C_{\min})$	xmax(Cap_min_measured_cont) - xmin(Cap_min_measured_cont)
$\Delta C(C_{\min})$	ymax(Cap_min_measured_cont, 0.1, 0.6) - ymin(Cap_min_measured_cont, 0.1, 0.6)
$d_{CV}(C_{\min})$	Delta_C_cap_min / Delta_Vdc_cap_min
$F_{\max}$	1 / (2 * 3.1416 * sqrt(236.20p * (Cap_min_measured + 208f + 114.5f)))
$F_{\max,cont}$	flatten(Freq_max, Vdc)
$K_{VCO} \text{ at } F_{\max}$	abs(derivative(Freq_max_cont))
$K_{VCO,max} \text{ at } F_{\max}$	ymax(o(KVCO_at_maxF, analysisName="ac"), 0.1, 0.6)

2. Testbench 2: Temperature Sweep Characterization

Testbench: Tese_Varactor_characterization_tb2

Sweep: Temperature sweep (temp) from -40°C to 125°C (11 points, step 16.5).

2.1 Minimum Frequency ($f_{\min} = 6\text{ GHz}$) / $C_{\max}$ Analysis

OUTPUT	EXPRESSION
I_{dev} (Device_current_M7)	i(/V24/PLUS)
Q-factor	abs(imag(Device_current_M7) / real(Device_current_M7))
$C_{\max,\text{meas}}$	abs(imag(Device_current_M7) / (2 * 3.14159 * 6e9))
$C_{\max,\text{cont}}$	flatten(Cap_max_measured, temp)
ΔT ($C_{\max}$)	xmax(Cap_max_measured_cont) - xmin(Cap_max_measured_cont)
ΔC ($C_{\max}$)	ymax(Cap_max_measured_cont, -40, 125) - ymin(Cap_max_measured_cont, -40, 125)
d_{CT} ($C_{\max}$)	Delta_C_cap_max / Delta_Temp_cap_max
$C_{\text{std},\text{max}}$	ymax(Cap_max_measured_cont, -40, 125) / Cap_max_measured_cont
% Error $C_{\text{std},\text{max}}$	(ymax(C_standard_cap_max, -40, 125) - ymin(C_standard_cap_max, -40, 125)) * 100

2.2 Maximum Frequency ($f_{\max} = 9.5\text{ GHz}$) / $C_{\min}$ Analysis

OUTPUT	EXPRESSION
I_{dev} (Device_current_M7)	i(/V24/PLUS)
Q-factor	abs(imag(Device_current_M7) / real(Device_current_M7))
$C_{\min,\text{meas}}$	abs(imag(Device_current_M7) / (2 * 3.14159 * 9.5e9))
$C_{\min,\text{cont}}$	flatten(Cap_min_measured, temp)
ΔT ($C_{\min}$)	xmax(Cap_min_measured_cont) - xmin(Cap_min_measured_cont)
ΔC ($C_{\min}$)	ymax(Cap_min_measured_cont, -40, 125) - ymin(Cap_min_measured_cont, -40, 125)
d_{CT} ($C_{\min}$)	Delta_C_cap_min / Delta_Temp_cap_min
$C_{\text{std},\text{min}}$	ymax(Cap_min_measured_cont, -40, 125) / Cap_min_measured_cont
% Error $C_{\text{std},\text{min}}$	(ymax(C_standard_cap_min, -40, 125) - ymin(C_standard_cap_min, -40, 125)) * 100